

Creating EC2 instance:

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name: project-1

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.10 AMI
ami-0bc7a3dc7f5b1e02a (64-bit x86), self-provisioned / ami-0bc3dc8b2c7694d5 (64-bit ARM), self-provisioned
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.10) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Summary

Number of instances: 1

Software image (AMI)
Amazon Linux 2023 AMI 2023.12...[read more](#)
ami-0bc7a3dc7f5b1e02a

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

Installing Docker :

```
[root@ip-172-31-36-53 ~]# systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-06-24 11:59:05 UTC; 31s ago
 TriggeredBy: ● docker.socket
   Docs: https://docs.docker.com
   Main PID: 29884 (dockerd)
     Tasks: 9
    Memory: 30.6M
       CPU: 338ms
    CGroup: /system.slice/docker.service
            └─29884 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768

Jun 24 11:59:05 ip-172-31-36-53.ap-south-1.compute.internal systemd[1]: Starting docker.service - Docker Application Co
Jun 24 11:59:05 ip-172-31-36-53.ap-south-1.compute.internal dockerd[29884]: time="2026-06-24T11:59:05.276082345Z" level=
Jun 24 11:59:05 ip-172-31-36-53.ap-south-1.compute.internal dockerd[29884]: time="2026-06-24T11:59:05.351006196Z" level=
Jun 24 11:59:05 ip-172-31-36-53.ap-south-1.compute.internal dockerd[29884]: time="2026-06-24T11:59:05.753887155Z" level=
Jun 24 11:59:05 ip-172-31-36-53.ap-south-1.compute.internal dockerd[29884]: time="2026-06-24T11:59:05.783051658Z" level=
Jun 24 11:59:05 ip-172-31-36-53.ap-south-1.compute.internal dockerd[29884]: time="2026-06-24T11:59:05.783207216Z" level=
Jun 24 11:59:05 ip-172-31-36-53.ap-south-1.compute.internal dockerd[29884]: time="2026-06-24T11:59:05.812629200Z" level=
Jun 24 11:59:05 ip-172-31-36-53.ap-south-1.compute.internal systemd[1]: Started docker.service - Docker Application Con
Lines 1-20/20 (END)
```

Installing Kubernetes:

```
root@ip-172-31-36-53:~
[root@ip-172-31-36-53 ~]# kubectl version --client
Client Version: v1.36.2
Kustomize Version: v5.8.1
[root@ip-172-31-36-53 ~]#
```

Clone Git repo:

```
[root@ip-172-31-36-53 ~]# mkdir project-1
[root@ip-172-31-36-53 ~]# cd project-1/
[root@ip-172-31-36-53 project-1]# git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 10 (from 1)
Receiving objects: 100% (11/11), 101.89 KiB | 218.00 KiB/s, done.
[root@ip-172-31-36-53 project-1]#
```

```

root@ip-172-31-36-53 project-1]# ls -ltrh
total 0
drwxr-xr-x. 4 root root 47 Jun 24 12:29 Brain-Tasks-App
[root@ip-172-31-36-53 project-1]# cd Brain-Tasks-App/
[root@ip-172-31-36-53 Brain-Tasks-App]# ls -ltrh
total 4.0K
-rw-r--r--. 1 root root 1.7K Jun 24 12:29 README.md
drwxr-xr-x. 3 root root 54 Jun 24 12:29 dist
[root@ip-172-31-36-53 Brain-Tasks-App]# cd dist/
[root@ip-172-31-36-53 dist]# ls -ltrh
total 8.0K
-rw-r--r--. 1 root root 1.5K Jun 24 12:29 vite.svg
-rw-r--r--. 1 root root 603 Jun 24 12:29 index.html
drwxr-xr-x. 2 root root 57 Jun 24 12:29 assets
[root@ip-172-31-36-53 dist]# |

```

Creating Dockerfile:

```

FROM nginx:alpine

COPY dist/ /usr/share/nginx/html

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]

```

Build Docker Image:

```

-rw-r--r-- 1 root root 100 Jun 24 12:46 Dockerfile
[root@ip-172-31-36-53 Brain-Tasks-App]# docker build -t brain-task-app:v1 .
[*] Building 1.2s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 197B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 586B
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:54f2a904c251d5a34adf545a72d32515a15e80418dae0266e23be2e18c66fefa
=> CACHED [2/2] COPY dist/ /usr/share/nginx/html
=> exporting to image
=> => exporting layers
=> => writing image sha256:19d2b84573d179d19c643ebda8b86fa74742125f600fa992aefbe67cf0ecee19
=> => naming to docker.io/library/brain-task-app:v1
[root@ip-172-31-36-53 Brain-Tasks-App]#

```

Doker container created:

```

REPOSITORY      TAG         IMAGE ID      CREATED        SIZE
brain-task-app  v1          7fa49fcd4fb9  4 seconds ago  62.5MB
[root@ip-172-31-36-53 Brain-Tasks-App]# docker run -d --name brain-app -p 3000:80 brain-task-app:v1
f4afd0e28a3c22be7da2cb003cf7230215b01d27cc78a28ff6d9b2d36f0800d5
[root@ip-172-31-36-53 Brain-Tasks-App]# docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
f4afd0e28a3c   brain-task-app:v1  "/docker-entrypoint..."  6 seconds ago  Up 5 seconds  0.0.0.0:3000->80/tcp, :::3000->80/tcp  brain-app
[root@ip-172-31-36-53 Brain-Tasks-App]# |

```

Enable Security Group:

The screenshot shows the AWS Management Console interface for a security group. A green notification banner at the top states: "Inbound security group rules successfully modified on security group (sg-0cab32ec4330283e8 | launch-wizard-16)". Below this, the security group details are displayed:

- Security group name:** launch-wizard-16
- Security group ID:** sg-0cab32ec4330283e8
- Description:** launch-wizard-16 created 2026-06-24T11:09:47.402Z
- VPC ID:** vpc-097050c1a56b9a7f1
- Owner:** 134943601221
- Inbound rules count:** 2 Permission entries
- Outbound rules count:** 1 Permission entry

Below the details, there are tabs for "Inbound rules", "Outbound rules", "Sharing", "VPC associations", "Related resources - new", and "Tags". The "Inbound rules" tab is selected, showing a table with 2 rules:

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-02fbc0af8a89ca274	IPv4	All TCP	TCP	0 - 65535	0.0.0.0/0	-
-	sg-0e855c9cc4ba0cd8	IPv4	SSH	TCP	22	0.0.0.0/0	-

brain task app running:

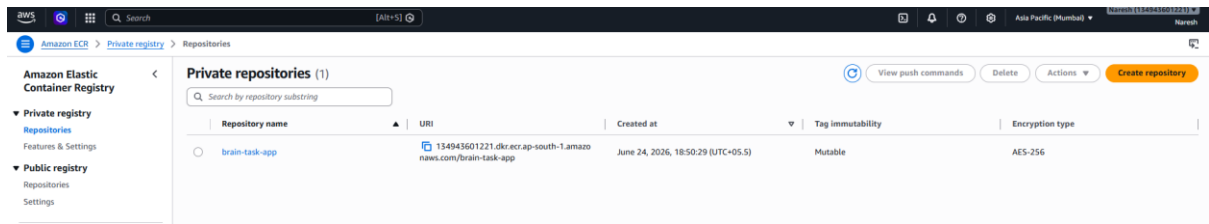
The screenshot shows the Brain Tasks application interface. At the top, there's a header with the text "Brain Tasks" and a subtitle "Organize your thoughts, simplify your life". A green button labeled "+ Create Task" is in the top right corner. Below the header is a search bar with the placeholder text "Search tasks...". Under the search bar are four tabs: "All Tasks", "Pending", "Completed", and "Priority". The "All Tasks" tab is selected. In the center of the page, there's a large white box with a document icon and the text "No tasks found! Create your first task to get started".

Creating ECR Repo:

The screenshot shows the AWS Management Console interface for creating a private repository. The page is titled "Create private repository". It has three main sections:

- General settings:** Includes a "Repository name" field with the value "brain-task-app". Below it, a note states: "14 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-./".
- Image tag settings:** Includes a "Image tag mutability" section with two options: "Mutable" (selected) and "Immutable". Below this is a "Mutable tag exclusions" section with a text input field and an "Add filter" button. A note states: "Filters must only contain letters, numbers, and special characters _-./-*. Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list."
- Encryption settings:** Includes a warning message: "The encryption settings for a repository can't be changed once the repository is created."

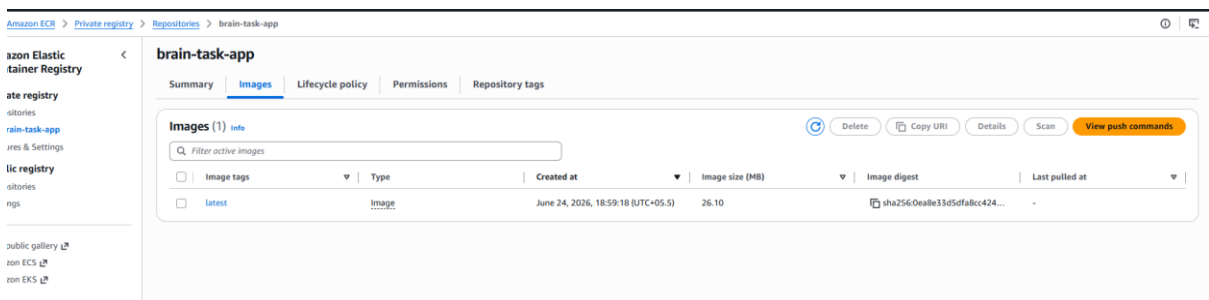
At the bottom, there's a section for "Encryption configuration" with a note: "By default, repositories use the industry standard Amazon Elastic File System (EFS) encryption. You can optionally choose to use a key stored in the AWS Key Management Service (KMS) to encrypt the content in your repository."



Docker image pushed into AWS ECR repo:

```
[root@ip-172-31-36-53 Brain-Tasks-App]# aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 134943601221.dkr.ecr.ap-south-1.amazonaws.com
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-172-31-36-53 Brain-Tasks-App]# docker build -t brain-task-app .
[+] Building 1.2s (7/7) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 197B                             0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine 1.1s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 566B                                  0.0s
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:54f2a984c251d5a34adf545a72d32515a15e88418dae8266e23be2e18c66fefa 0.8s
=> CACHED [2/2] COPY dist/ /usr/share/nginx/html                0.8s
=> exporting to image                                           0.8s
=> => exporting layers                                           0.8s
=> => writing image sha256:7fa49fcd4fb9459424c844665ff72a70a8a1a541cae7c163951138b05aa5ac12 0.0s
=> => naming to docker.io/library/brain-task-app                0.0s
[root@ip-172-31-36-53 Brain-Tasks-App]# docker tag brain-task-app:latest 134943601221.dkr.ecr.ap-south-1.amazonaws.com/brain-task-app:latest
[root@ip-172-31-36-53 Brain-Tasks-App]# docker push 134943601221.dkr.ecr.ap-south-1.amazonaws.com/brain-task-app:latest
The push refers to repository [134943601221.dkr.ecr.ap-south-1.amazonaws.com/brain-task-app]
155d7206f047: Pushed
afb66a24dc8: Pushed
4979493d808a: Pushed
1876c8091457: Pushed
d32a1b6bc05e: Pushed
3c4c14c5eb84: Pushed
c54525213cd0: Pushed
c91b7a48dd9f: Pushed
31ad4a471c68: Pushed
latest: digest: sha256:0ea8e33d5dfa8cc424ee43696ea7289271cadd8b5d07e48f5e1d64defc9c9eaf size: 2199
[root@ip-172-31-36-53 Brain-Tasks-App]#
```



Creating EKS cluster:

```
[root@ip-172-31-36-53 ~]# eksctl create cluster --name brain-task-cluster --region ap-south-1 --nodegroup-name standard-nodes --node-type t3.medium --nodes 2
2026-06-24 13:53:57 [I] eksctl version 0.227.0
2026-06-24 13:53:57 [I] using region ap-south-1
2026-06-24 13:53:57 [I] setting availability zones to [ap-south-1b ap-south-1c ap-south-1a]
2026-06-24 13:53:57 [I] subnets for ap-south-1b - public:192.168.0.0/19 private:192.168.96.0/19
2026-06-24 13:53:57 [I] subnets for ap-south-1c - public:192.168.32.0/19 private:192.168.128.0/19
2026-06-24 13:53:57 [I] subnets for ap-south-1a - public:192.168.64.0/19 private:192.168.160.0/19
2026-06-24 13:53:57 [I] nodegroup "standard-nodes" will use "" [AmazonLinux2023/1.34]
2026-06-24 13:53:57 [I] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-06-24 13:53:57 [I] using Kubernetes version 1.34
2026-06-24 13:53:57 [I] creating EKS cluster "brain-task-cluster" in "ap-south-1" region with managed nodes
2026-06-24 13:53:57 [I] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-06-24 13:53:57 [I] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=ap-south-1 --cluster=brain-task-cluster'
2026-06-24 13:53:57 [I] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "brain-task-cluster" in "ap-south-1"
2026-06-24 13:53:57 [I] CloudWatch logging will not be enabled for cluster "brain-task-cluster" in "ap-south-1"
2026-06-24 13:53:57 [I] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=ap-south-1 --cluster=brain-task-cluster'
2026-06-24 13:53:57 [I] default addons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS addons
2026-06-24 13:53:57 [I] 2 sequential tasks: { create cluster control plane "brain-task-cluster",
2 sequential sub-tasks: {
2 sequential sub-tasks: {
1 task: { create addons },
wait for control plane to become ready,
},
create managed nodegroup "standard-nodes",
}
}
2026-06-24 13:53:57 [I] building cluster stack "eksctl-brain-task-cluster-cluster"
2026-06-24 13:53:57 [I] deploying stack "eksctl-brain-task-cluster-cluster"
```

```

2026-06-24 14:53:14 [I] waiting for the control plane to become ready
2026-06-24 14:53:15 [✓] saved kubeconfig as "/root/.kube/config"
2026-06-24 14:53:15 [I] no tasks
2026-06-24 14:53:15 [✓] all EKS cluster resources for "brain-task-cluster" have been created
2026-06-24 14:53:15 [I] nodegroup "ng-a4628da9" has 2 node(s)
2026-06-24 14:53:15 [I] node "ip-192-168-30-165.ap-south-1.compute.internal" is ready
2026-06-24 14:53:15 [I] node "ip-192-168-77-57.ap-south-1.compute.internal" is ready
2026-06-24 14:53:15 [I] waiting for at least 2 node(s) to become ready in "ng-a4628da9"
2026-06-24 14:53:15 [I] nodegroup "ng-a4628da9" has 2 node(s)
2026-06-24 14:53:15 [I] node "ip-192-168-30-165.ap-south-1.compute.internal" is ready
2026-06-24 14:53:15 [I] node "ip-192-168-77-57.ap-south-1.compute.internal" is ready
2026-06-24 14:53:15 [✓] created 1 managed nodegroup(s) in cluster "brain-task-cluster"
2026-06-24 14:53:15 [I] creating addon: metrics-server
2026-06-24 14:53:15 [I] successfully created addon: metrics-server
2026-06-24 14:53:16 [I] kubectrl command should work with "/root/.kube/config", try 'kubectl get nodes'
2026-06-24 14:53:16 [✓] EKS cluster "brain-task-cluster" in "ap-south-1" region is ready
[root@ip-172-31-36-53 ~]#

```

Cluster created:

The screenshot shows the Amazon Elastic Kubernetes Service (EKS) console. On the left, there's a sidebar with 'Amazon Elastic Kubernetes Service' and 'Clusters'. The main area is titled 'Clusters (1)' and shows a table with one cluster, 'brain-task-cluster', which is 'Active'. The table includes columns for Cluster name, Status, Kubernetes version (1.34), Support period, Upgrade policy, Created (13 minutes ago), and Provider (EKS). There are buttons for 'Delete' and 'Create cluster' at the top right.

Cluster name	Status	Kubernetes version	Support period	Upgrade policy	Created	Provider
brain-task-cluster	Active	1.34 Upgrade now	Standard support until December 2, 2026	Extended support	13 minutes ago	EKS

Deployment.yml file:

```

root@ip-172-31-36-53:~/proj1
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-app
  labels:
    app: brain-tasks-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks-app
  template:
    metadata:
      labels:
        app: brain-tasks-app
    spec:
      containers:
        - name: brain-tasks-app
          image: 134943601221.dkr.ecr.ap-south-1.amazonaws.com/brain-task-app:latest
          ports:
            - containerPort: 3000

```

Service.yml file:

```

root@ip-172-31-36-53:~/proj1
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  ports:
    - port: 80
      targetPort: 3000
  selector:
    app: brain-tasks-app

```

Buildspec.yml:

```
version: 0.2

phases:

  pre_build:
    commands:
      - echo Logging in to Amazon ECR...
      - aws ecr get-login-password --region $AWS_REGION | docker login --username AWS --password-stdin $AWS_ACCOUNT_ID.dkr.ecr.$AWS_REGION.amazonaws.com
      - echo Setting up Kubeconfig...
      - aws eks update-kubeconfig --region $AWS_REGION --name $EKS_CLUSTER_NAME

  build:
    commands:
      - echo Build started on `date`
      - echo Building the Docker image...
      - docker build -t brain-tasks-app .
      - docker tag brain-tasks-app:latest $REPOSITORY_URI:$CODEBUILD_RESOLVED_SOURCE_VERSION
      - docker tag brain-tasks-app:latest $REPOSITORY_URI:latest

  post_build:
    commands:
      - echo Build completed on `date`
      - echo Pushing the Docker image to ECR...
      - docker push $REPOSITORY_URI:$CODEBUILD_RESOLVED_SOURCE_VERSION
      - docker push $REPOSITORY_URI:latest
      - echo Updating Kubernetes manifests...
      - sed -i "s|brain-tasks-app:latest|brain-tasks-app:$CODEBUILD_RESOLVED_SOURCE_VERSION|g" kubernetes/deployment.yml
      - echo Deploying to AWS EKS cluster...
      - kubectl apply -f kubernetes/deployment.yml
      - kubectl apply -f kubernetes/service.yml
```

GitHub Repo created and codes are pushed:

The screenshot shows the GitHub repository page for 'Brain-task-app'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The 'Code' tab is selected, showing a list of files: root, dist, kubernetes, 1, Dockerfile, README.md, and buildspec.yml. The 'README' tab is also visible, showing the text 'DevOps Practice Project – Dist Directory'.

Creating Codebuild project:

The screenshot shows the AWS CodeBuild 'Create build project' wizard. The 'Project configuration' section is filled out with 'Brain-task-app-Codebuild' as the project name. The 'Project type' is set to 'Default project'. The 'Source' section shows 'Source 1 - Primary' with 'Source provider' set to 'GitHub'. The 'Repository' is set to 'Repository in my GitHub account' with the URL 'https://github.com/Naresh04/brain-task-app'.

Added Source GitHub Repo:

▼ Source

Add source

Source 1 - Primary

Source provider

GitHub

Credential

✔ Your account is successfully connected by using an AWS managed GitHub App. Manage account credentials.

☐ Use override credentials for this project only

Repository

☒ Repository in my GitHub account

☐ Public repository

☐ GitHub scoped webhook

Repository

Q https://github.com/Nareshy04/Brain-task-app

X

🔄

Source version - optional info

Enter a pull request, branch, commit ID, tag, or reference and a commit ID.

▶ Additional configuration

Git clone depth, Git submodules, Build status config

Added Builds.yml file:

▼ Builds

Build specifications

☐ Insert build commands

☒ Use a builds.yml file

Store build commands in a YAML-formatted builds.yml file

Buildspec name - optional

By default, CodeBuild looks for a file named builds.yml in the source code root directory. If your builds.yml file uses a different name or location, enter its path from the source root here (for example, buildspec-two.yml or configuration/buildspec.yml).

buildspec.yml

▼ Batch configuration

You can run a group of builds as a single execution. Batch configuration is also available in advanced option when starting build.

☐ Define batch configuration - optional

You can also define or override batch configuration when starting a build batch.

▼ Artifacts

Add artifact

Artifact 1 - Primary

Type

CodeBuild Project created:

Developer Tools

CodeBuild

► Source • CodeCommit

► Artifacts • CodeArtifact

▼ Build • CodeBuild

Getting started

Build projects

Build project

Settings

Build history

Report groups

Report history

Compute fleets

Account metrics

▼ Related integrations

Jenkins

GitHub Actions

GitLab runners

Project created

You have successfully created the following project: Brain-task-app-Codebuild

Create a notification rule for this project

Brain-task-app-Codebuild

Actions

Create trigger

Edit

Clone

Debug build

Start build with overrides

Start build

Configuration

Source provider

GitHub

Primary repository

Nareshy04/brain-task-app

Artifacts upload location

-

Service role

arn:aws:iam::134943601221:role/service-role/codebuild-Brain-task-app-Codebuild-service-role

Build history

Batch history

Project details

Build triggers

Metrics

Debug sessions

Build history

Stop build

View artifacts

View logs

Delete builds

Retry build

Debug build

Build run

Status

Build number

Source version

Submitter

Duration

Completed

No results

There are no results to display.

Attach policy to access ECR :

Amazon Elastic Kubernetes Service

Clusters

brain-task-cluster

on Elastic Service

ard

s

gs

ard settings

ettings

on EKS Anywhere

se Subscriptions

d services

ECR

hch

IAM access entries (5)

Find access entry by property filtering

View details

Edit

Delete

Create

	IAM principal ARN	Type	Username	Group names	Access policies
☐	arn:aws:iam::134943601221:role/aws-service-role/eks.amazonaws.com/AWSIServiceRoleforAmazonEKS	Standard	arn:aws:sts::134943601221:assumed-role/AWSIServiceRoleforAmazonEKS/[SessionName]	eks:managed	AmazonEKSClusterInsightsPolicy, AmazonEKSEventPolicy
☐	arn:aws:iam::134943601221:role/eksctl-brain-task-cluster-nodegroup-NodeInstanceRole-YZMOYVSHA91W	EC2 Linux	system:node:[EC2PrivateDNSName]	system:nodes	
☐	arn:aws:iam::134943601221:role/service-role/codebuild-Brain-task-app-Codebuild-service-role	Standard	arn:aws:sts::134943601221:assumed-role/codebuild-Brain-task-app-Codebuild-service-role/[SessionName]		AmazonEKSClusterAdminPolicy
☐	arn:aws:iam::134943601221:root	Standard	arn:aws:iam::134943601221:root		AmazonEKSClusterAdminPolicy
☐	arn:aws:iam::134943601221:user/EKS-user	Standard	arn:aws:iam::134943601221:user/EKS-user		AmazonEKSClusterAdminPolicy

Creating Pipeline:

The screenshot shows the 'Choose creation option' step of the AWS CodePipeline console. The left sidebar lists steps 1 through 6. The main content area is titled 'Choose creation option' and 'Step 1 of 7'. It features a 'Category' section with three radio buttons: 'Deployment', 'Continuous Integration', and 'Automation'. The 'Build custom pipeline' option is selected and highlighted with a blue border. At the bottom right, there are 'Cancel' and 'Next' buttons.

Added Pipeline name:

The screenshot shows the 'Choose pipeline settings' step of the AWS CodePipeline console. The left sidebar lists steps 1 through 7. The main content area is titled 'Choose pipeline settings' and 'Step 2 of 7'. It features a 'Pipeline settings' section with a 'Pipeline name' text input field containing 'brain-tasks-pipeline'. Below this is the 'Execution mode' section with three radio buttons: 'Superseded', 'Queued' (selected), and 'Parallel'. The 'Service role' section has two options: 'New service role' (selected) and 'Existing service role'. Below this is the 'Role name' text input field containing 'AWSCodePipelineServiceRole-ap-south-1-brain-tasks-pipeline'. At the bottom, there is an 'Advanced settings' section with a plus icon and a description: 'Configure artifact store location, encryption settings, and pipeline variables for your pipeline.'

Source file added:

The screenshot shows the 'Add source stage' step of the AWS CodePipeline console. The left sidebar lists steps 1 through 7. The main content area is titled 'Add source stage' and 'Step 3 of 7'. It features a 'Source' section with a 'Source provider' dropdown menu set to 'GitHub (via GitHub App)'. Below this is the 'Connection' section with a text input field containing 'arn:aws:codeconnections:ap-south-1:...' and a 'Connect to GitHub' button. The 'Repository name' section has a text input field containing 'Nareshvj04/Brain-task-app'. The 'Default branch' section has a text input field containing 'main'. The 'Output artifact format' section has two radio buttons: 'CodePipeline default' (selected) and 'Full clone'. At the bottom, there is a checkbox labeled 'Enable automatic retrv on staoe failure' which is checked.

Build run succeeded:

The screenshot shows the AWS CodeBuild console for the 'Brain-task-app-Codebuild' project. The configuration section shows the source provider as GitHub, primary repository as 'Naresh04/brain-task-app', and artifacts upload location as '-'. The build history section shows a single build run that succeeded.

Build run	Status	Build number	Source version	Submitter	Duration	Completed
Brain-task-app-Codebuild:9727991-ba2a-4e99-87bc-1368732073e9	Succeeded	22	-	root	44 seconds	3 minutes ago

```
176 [Container] 2026/06/25 12:10:28.354948 Running command echo Updating Kubernetes manifests...
177 [Container] 2026/06/25 12:10:28.362715 Running command sed -i "s|brain-tasks-app:latest|brain-tasks-app:$CODEBUILD_RESOLVED_SOURCE_VERSION|g" kubernetes/deployment.yml
178 [Container] 2026/06/25 12:10:28.402770 Running command echo Deploying to AWS EKS cluster...
179 [Container] 2026/06/25 12:10:28.410346 Running command kubectl apply -f kubernetes/deployment.yml
180 [Container] 2026/06/25 12:10:32.259171 Running command kubectl apply -f kubernetes/service.yml
181 [Container] 2026/06/25 12:10:33.172185 Phase complete: POST_BUILD State: SUCCEEDED
182 [Container] 2026/06/25 12:10:33.172295 Phase context status code: Message:
183 [Container] 2026/06/25 12:10:33.219735 Set report auto-discover timeout to 5 seconds
184 [Container] 2026/06/25 12:10:33.219779 Expanding base directory path: .
185 [Container] 2026/06/25 12:10:33.223039 Assembling file list
186 [Container] 2026/06/25 12:10:33.223046 Expanding .
187 [Container] 2026/06/25 12:10:33.226296 Expanding file paths for base directory .
188 [Container] 2026/06/25 12:10:33.226311 Assembling file list
189 [Container] 2026/06/25 12:10:33.226314 Expanding **/*
190 [Container] 2026/06/25 12:10:33.230092 No matching auto-discover report paths found
191 [Container] 2026/06/25 12:10:33.230023 Report auto-discover file discovery took 0.010288 seconds
192 [Container] 2026/06/25 12:10:33.230031 Phase complete: UPLOAD_ARTIFACTS State: SUCCEEDED
193 [Container] 2026/06/25 12:10:33.230041 Phase context status code: Message:
```

Image pushed to ECR:

The screenshot shows the AWS ECR console for the 'brain-task-app' repository. It lists three images pushed to the repository.

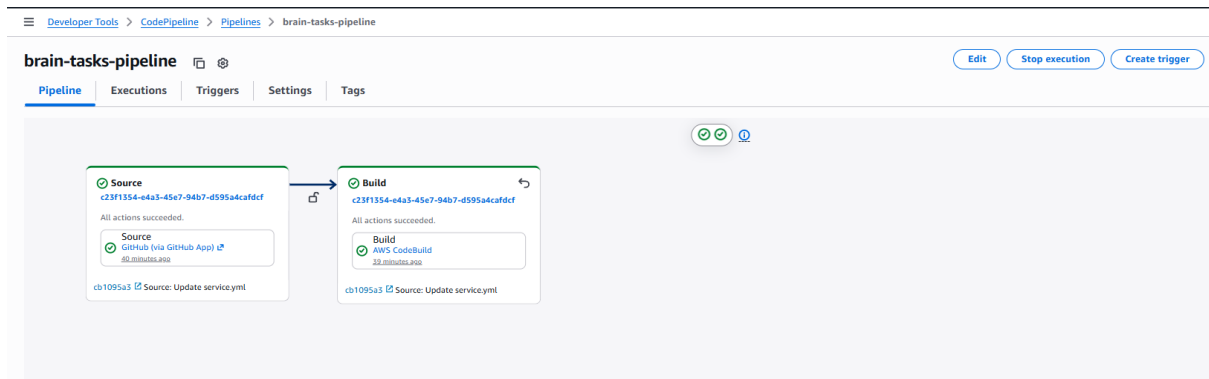
Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
cb1095a32c80c4239f8b8179651974a0042f277, latest	image	June 25, 2026, 18:17:24 (UTC+05.5)	26.10	sha256:15e2815f931f06eac...	-
-	image	June 25, 2026, 18:17:20 (UTC+05.5)	26.10	sha256:14a7cac29059c0421...	-
6a5c98361a852104fa55efac72071555d0e6ad9	image	June 25, 2026, 18:06:56 (UTC+05.5)	26.10	sha256:3a796521f2cb2511d...	June 25, 2026, 18:14:50 (UTC+05.5)

Build Project Succeeded:

The screenshot shows the AWS CodeBuild console for the 'Brain-task-app-Codebuild' project. The build status is 'Succeeded'.

Name	Source provider	Repository	Latest build status	Description	Last Modified
Brain-task-app-Codebuild	GitHub	Naresh04/brain-task-app	Succeeded	-	53 minutes ago

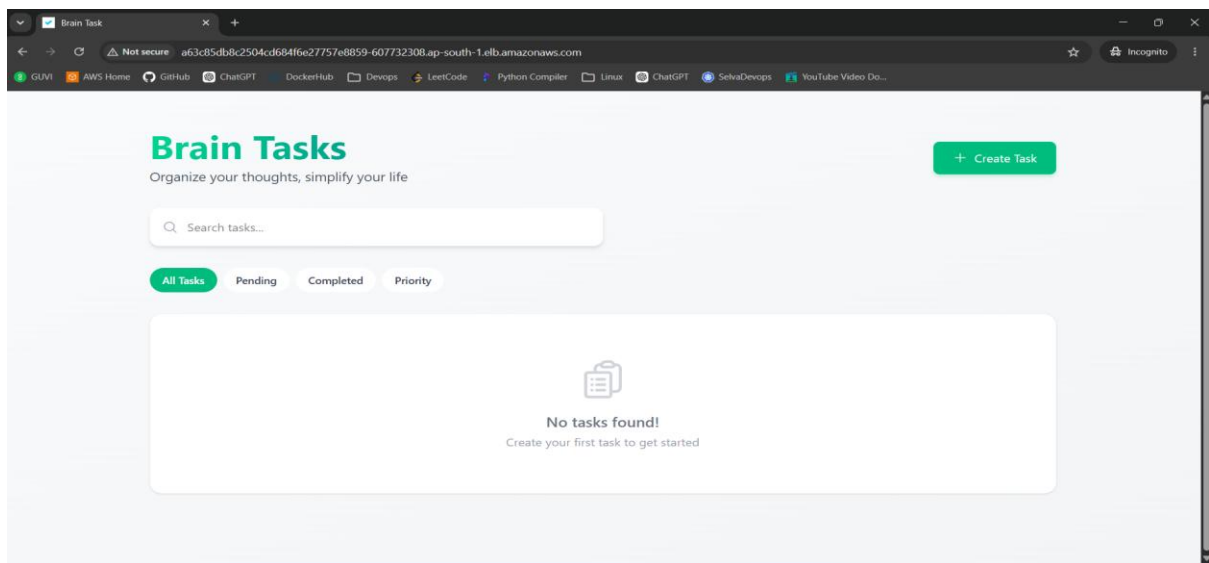
CodePipeline :



Pods and Service running:

```
[root@ip-172-31-36-53 ~]# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
brain-tasks-app-6f859b85bf-82hz8    1/1     Running   0           2m49s
brain-tasks-app-6f859b85bf-bp84w    1/1     Running   0           2m52s
[root@ip-172-31-36-53 ~]# kubectl get svc
NAME      TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
brain-tasks-service  LoadBalancer  10.100.18.36  a63c85db8c2504cd684f6e27757e8859-607732308.ap-south-1.elb.amazonaws.com  80:31806/TCP,443/TCP  23s
kubernetes  ClusterIP     10.100.0.1    <none>        <none>            99m
```

Using External IP access to website:



Monitoring Cloudwatch dashboard created:

